David Joy

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Education

University of Georgia(UGA) **Athens,USA**

Major: Artificial Intelligence -Thesis Track, GPA: 3.82/4.00 ***Jan 2024 – Aug 2026***

- Relevant Coursework: Artificial Intelligence, Data Mining, Privacy-Preserving Data Analysis,Computer Vision

National Institute of Technology Calicut (NITC) **Kozhikode, India**

Major: Electronics and Communication Engineering, CGPA: 7.64/10 ***Jul 2019 – May 2023***

- Relevant Coursework: Probability & Statistics, C++ Programming, Python Programming, Data Structures & Algorithms, Digital Signal Processing

Experience

MODEL Lab - University of Georgia **Mar 2024 – Present**

Research Assistant *Athens,Georgia*

- Research Assistant at the **Manufacturing Optimization Design Engineering-Education Lab(MODEL) under Dr. Beshoy Morkos** in developing computer vision and LLM AI models to accelerate smart manufacturing.
- Set up **end-to-end pipelines** to develop **intelligent machines** in the manufacturing sector for **Middleby** industrial equipments that utilizes computer vision and image segmentation to **increase yield by 50%**.
- Engineering team **member of the Georgia - AI for Manufacturing (GA-AIM) project** funded by the U.S Economic Development Administration (EDA) in creating scalable local ML applications for **100+ manufacturers**

Indian Institute of Science **May 2023 – Dec 2023**

Machine Learning Researcher *Bengaluru,Karnataka*

- Machine Learning Researcher in the **web applications team** of the **Design and Innovation Centre** at IISc. Bangalore.
- Incorporated **automated video segmenting** using **computer vision** on video screencast recordings of collaborative videos
- Implemented **Siamese Neural Networks** for **transfer learning** for video action recognition tasks that replaced manual labeling.

Projects

Mini-Inference | *Python, CUDA C++, Triton, PyTorch*| **Sep 2026 - Oct 2026**

- Built an **LLM inference engine** for MoE (OLMoE),Qwen3 models, with **continuous batching, paged KV cache, prefix caching, chunked prefill** and **FP8/INT8/INT4 quantization** mimicking serving engines **SGLang** and **vLLM**.
- **Raised throughput 2.3×** (1,521 to 3,447 tok/s) using profile-driven optimization: **fused-MoE Triton and CUDA C++ kernels**, **a paged-attention kernel at 92% of peak bandwidth**, and CUDA graphs.
- Profiled a dense-model decode bottleneck to cuBLAS kernel selection with a custom GEMM benchmark fused projections with autotuned weight layouts, **saving 2 ms per step** as predicted.

FlashDecode-CUDA | *CUDA C++ , Nsight Compute,PyTorch , Python*| **Aug 2026 - Oct 2026**

- Designed a **paged GQA decode-attention CUDA kernel** for RTX 5090 GPU **achieving 6.2–6.7x faster** than PyTorch Attention implementation at working sets beyond L2, via **split-KV decomposition and 16-byte vectorized loads**.
- Built a Qwen2.5-1.5B decode engine with a **paged KV cache** and **fused RMSNorm kernel**, validated bit-exact against HuggingFace across all 28 layers.

Kiln-Compiler | *Python, MLIR, Triton, CUDA*| **Oct 2026 - Nov 2026**

- **Built a machine learning compiler** that achieves comparable operational performance on Python relative to baseline PyTorch.
- Achieved **16% more TFLOPs** on an RTX 5090 with **autotuned Triton matmuls**; a GPT-2 block runs **1.2–1.9x faster** than PyTorch, matching Pytorch's compile modules.

Publications

- A Comparative Study Between Humans and GPT for Tradespace Analysis - ASME IDETC 2025

Technical Skills

Languages and Skills: Python, C++, SQL, Machine Learning, R, Computer Vision
Developer Tools: Linux bash, Git, Docker, ONNX, AWS, Azure,vLLM, SGLang
Technologies/Frameworks: PyTorch, Tensorflow, CUDA, MLIR, LLVM